Lecture 5: Multiple regression and the referee report

Empirical Methods for Law and Economics | EDLE

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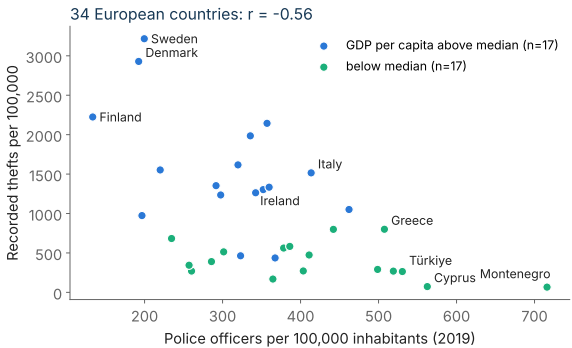
Today's questions

Do countries with more police record less theft
once we compare countries with similar income?

How do you write a referee report?

- By the end of today you will be able to:
  - fit a regression with several explanatory variables and read each coefficient "holding the others fixed";
  - predict how a coefficient changes when a variable is added or left out;
  - use indicators for more than two groups;
  - structure a referee report and tell a useful comment from a weak one.

DISCUSS Back to police and theft



- Lecture 3: $r = -0.56$ across 34 countries.
- Richer countries record more theft *and* employ fewer police per inhabitant.

Questions:

1. Why does income make the correlation hard to read?
2. With the data we have, how could we compare countries "with similar income"?

ANSWERS Back to police and theft

- Income is related to both variables: it can produce a negative correlation between police and recorded theft even if police had no effect on theft.

- Comparing countries with similar income:
  - split the sample by income and look within each group (few countries per group);
  - or let a regression compare them for us, by adding income as a second explanatory variable.

- The second option is **multiple regression**: today's tool.

Multiple regression

$$\text{theft}_i = \beta_0 + \beta_1 \,\text{police}_i + \beta_2 \ln(\text{GDP p.c.})_i + u_i$$

- OLS chooses $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ to minimise the sum of squared residuals, as with one variable.
- $\hat{\beta}_1$: difference in recorded theft associated with one more police officer per 100,000, **among countries with the same income**.
- "Holding income fixed" is a statistical comparison, not an experiment: nobody changes police while freezing income.
- Each coefficient answers a different question from the simple regression slope. Neither is "the" effect of police.

PREDICT What happens to the police coefficient?

- Simple regression: -3.64 recorded thefts per 100,000 for each additional police officer per 100,000.
- Now we add ln(GDP p.c.) as a second explanatory variable.

Predict, and write down your answer:

1. Will the police coefficient become larger in absolute value, smaller, or change sign?
2. Why? Use the two facts on the previous slides.

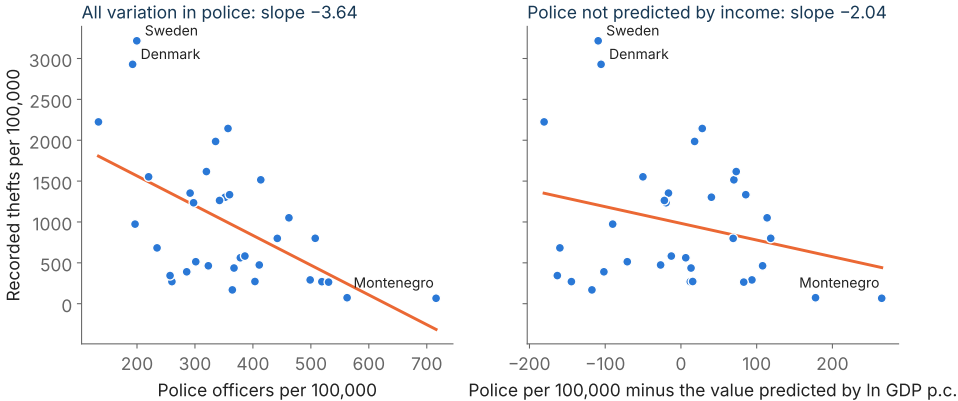
STATA Guided Stata 1: adding a control

```
use "$course_data/justice_europe_2019.dta", clear
generate ln_gdppc = ln(gdp_pc_usd)
regress theft_100k police_100k
estimates store simple
regress theft_100k police_100k ln_gdppc
estimates store multiple
estimates table simple multiple, b(%9.2f) se(%9.2f) stats(N r2)
```

Recorded thefts per 100,000	simple	multiple
Police per 100,000	−3.64 (0.96)	−2.04 (1.03)
ln GDP per capita		439.1 (153.3)
N / R^2	34 / 0.31	34 / 0.45

- The police coefficient shrinks by 44%; its 95% interval now includes zero (−4.15 to 0.07; $p = 0.057$).

What “holding income fixed” does



- Right panel: police minus the value predicted by income. The slope of theft on this “income-free” part of police is exactly the multiple-regression coefficient, −2.04.

STATA Guided Stata 2: the same coefficient in two steps

```
regress police_100k ln_gdppc
predict police_res, residuals
regress theft_100k police_res
```

- Step 1: predict police from income; `police_res` is the part of police that income does not predict.
- Step 2: regress theft on that part. Slope: -2.04, the same as in the multiple regression.
- The standard error of step 2 is not the right one (1.33 instead of 1.03): use the multiple regression for inference.
- Every coefficient in a multiple regression is built this way: it uses only the variation in that variable that the other explanatory variables do not predict.

Why the coefficient moved: omitted variable bias

$$\underbrace{\hat{\beta}_1^{\text{simple}}}_{-3.64} = \underbrace{\hat{\beta}_1^{\text{multiple}}}_{-2.04} + \underbrace{\hat{\beta}_2}_{439.1} \times \underbrace{\hat{\delta}}_{-0.00364}$$

- $\hat{\beta}_2$: income and theft, holding police fixed (positive: richer countries record more theft).
- $\hat{\delta}$: slope of ln(GDP p.c.) on police (negative: more police, lower income).
- Product: $439.1 \times (-0.00364) = -1.60$. Leaving income out pushes the police coefficient down by 1.60.
- Sign rule: the bias has the sign of (effect of the omitted variable) $\times$ (its relation with the included variable).
- The formula holds exactly for an omitted variable we observe. For variables we do not observe, we can only reason about the signs.

DISCUSS Which controls would you add?

- Outcome: recorded thefts per 100,000. Variable of interest: police per 100,000. Candidate controls:

1. share of the population living in cities;
2. unemployment rate;
3. recorded burglaries per 100,000;
4. share of households with home insurance;
5. police budget per inhabitant;
6. the first letter of the country's name.

For each one: does it belong in the regression? Why?

- Hint: ask whether it affects theft, whether it is related to police, and whether it is itself shaped by police or by the same recording process.

ANSWERS Which controls would you add?

- **Confounders** (affect recorded theft and are related to police): urbanisation (1), unemployment (2), insurance (4), which drives reporting. Add them if measured.
- **Bad control** (3): recorded burglaries come from the same reporting and recording process as recorded thefts, and police may affect them too. Holding them fixed removes part of what we want to measure.
- **Same variable twice** (5): the police budget mostly measures police again; with both in the model, neither coefficient has a clear meaning and both are imprecise.
- **Irrelevant** (6): unrelated to both; it adds noise, not bias.
- Rule: a good control is determined *before* the variable of interest and affects the outcome. More controls is not always better.

DISCUSS "−2.04: so police deter theft"

What would a referee say? Give at least two reasons why -2.04 is still not the effect of police.

- Use the three alternative explanations of Lecture 3: reverse causality, confounding, measurement.

ANSWERS "−2.04: so police deter theft"

- **Confounding**: income is only one confounder; urbanisation, unemployment and insurance are not in the model.
- **Reverse causality**: governments may hire police where theft is high. This pushes the coefficient *up*, so the true deterrent effect could be larger in absolute value.
- **Measurement**: more police can mean more thefts *recorded*. This also pushes the coefficient up.
- Chance: $p = 0.057$ with 34 countries.
- Conclusion: holding income fixed weakens the association, and the remaining coefficient is not an effect.

From two groups to four

- Lecture 4: a 0/1 regressor compares two groups (high income or not).
- The World Bank classifies economies into **four** income groups. One indicator per group, except one: the **base** (reference) group.

$$\text{time}_i = \beta_0 + \beta_1 \,\text{LowerMiddle}_i + \beta_2 \,\text{UpperMiddle}_i + \beta_3 \,\text{High}_i + u_i$$

- β_0: mean time in the base group (low income).
- $\beta_1, \beta_2, \beta_3$: difference between each group's mean and the base group's mean.
- All four indicators together would be perfectly collinear with the constant: Stata drops one.

STATA Guided Stata 3: indicators for four groups

```
use "$course_data/contracts_2019.dta", clear
generate byte income4 = 1 if income_group == "Low income"
replace income4 = 2 if income_group == "Lower middle income"
replace income4 = 3 if income_group == "Upper middle income"
replace income4 = 4 if income_group == "High income"
label define inc4 1 "Low" 2 "Lower middle" 3 "Upper middle" 4 "High"
label values income4 inc4
tabstat time_days, by(income4) statistics(n mean)
regress time_days i.income4
```

Group	*n*	Mean days
Low	30	680.4
Lower middle	47	684.4
Upper middle	55	633.8
High	59	621.3

`regress`	Coef.	(SE)
Lower middle	4.0	(71.4)
Upper middle	−46.7	(69.3)
High	−59.2	(68.5)
Constant	680.4	(55.8)

DISCUSS Reading indicators

Questions:

1. Write the coefficient on *High* (-59.2) as a sentence. Compared with which economies?
2. Which coefficients change if we use *High* as the base group (`regress time_days ib4.income4`)? Which numbers do not change?
3. None of the three differences is statistically significant. Does this mean income groups do not differ?

ANSWERS Reading indicators

- High-income economies take on average 59.2 days less than **low-income** economies (680.4 vs 621.3 days).

- With *High* as base: the constant becomes 621.3 and every coefficient is a difference from the high-income mean (Low: $+59.2$). The four group means, the fitted values and R^2 (0.008) do not change.

- No: the intervals are wide (High: -194 to $+76$ days). The data are compatible with no difference and with large ones.

DISCUSS Back to Voigt and Wulf, column (1)

- English legal origin: 0.24 (SE 0.09). The column also contains de jure independence, press freedom, polity, French legal origin and GDP per capita.
- Reference group: countries of socialist, German or Scandinavian legal origin.
- Footnote 11: press freedom and GDP per capita are correlated ($r = 0.71$). Press freedom has $p = 0.115$ in model 4, which includes GDP per capita, and $p < 0.001$ when GDP per capita is removed.

Questions:

1. Write the English coefficient as a sentence, now that you know what "holding fixed" means.
2. Why does the p-value of press freedom change so much when GDP per capita is removed?

Voigt, S. and Wulf, A.J. (2019), *Journal of Institutional Economics* 15(1): 99–120, Table 2 and footnote 11.

ANSWERS Back to Voigt and Wulf, column (1)

- Among countries with the same de jure independence, press freedom, polity score and GDP per capita, those of English legal origin score on average 0.24 points higher on the 0–1 de facto index than those of socialist, German or Scandinavian legal origin.

- With both variables in the model, the press freedom coefficient uses only the part of press freedom not predicted by GDP (and the other variables). When the two move together, that part is small: the estimate is imprecise.

- Removing GDP per capita gives press freedom the variation it shares with income: the coefficient becomes precise, but it now also picks up income.

The referee report: who reads it

- The **editor** decides; the referee advises.
- The referee has two goals: assess whether the paper is suitable for publication, and help the authors improve it (Thomson 2011, p. 151).
- Two documents:
  - the **report**, sent to the authors (anonymously);
  - the **letter to the editor**, confidential.
- In this course you write the report only: 1,500–2,000 words on the paper or seminar you chose.

Thomson, W. (2011), *A Guide for the Young Economist*, 2nd ed., MIT Press, Chapter 4.

DISCUSS Berk, Harvey and Hirshleifer (2017)

- Former editors of six leading journals were asked what goes wrong in refereeing. Their answer: referees "tend to make similar, correctable, mistakes" (p. 232).

From your reading:

1. Which three central problems of refereeing do the authors identify?
2. Why do they call the failure to separate the essential from the suggested "the highest-cost mistake"?
3. "Flaws and all, would I have been pleased to have written such a paper?" What is this question for?

Berk, J.B., Harvey, C.R. and Hirshleifer, D. (2017), *Journal of Economic Perspectives* 31(1): 231–244.

ANSWERS Berk, Harvey and Hirshleifer (2017)

- Three problems (p. 232): weighing the **importance** of a paper, without being distracted by minor flaws; separating what is **essential** for publication from what is a **suggestion**; respecting the **implicit contract** of a revise-and-resubmit.

- Referees want to look smart and inflate "minor blemishes to the status of major flaws" (p. 231). Unless secondary comments are labelled as such, they are "inherently coercive" (p. 236): authors cannot afford to ignore them, and papers grow longer without getting better.

- Importance first: all papers have flaws; the question is whether the flaws "actually invalidate the contribution" (p. 233).

- An essential comment needs a "scientifically based argument" (p. 235). A hunch that results do not pass the "smell test" is not one (pp. 235–236).

Anatomy of a report

1. **Summary**, in your own words: question, data and unit of observation, method, main findings; the main contribution; what you checked yourself (a table recomputed, the replication files).
2. **Main comments**: the problems that must be addressed, numbered. Each one:
   - a title that states the problem;
   - the issue, citing the exact table, column or page;
   - why it matters for a specific conclusion of the paper;
   - what would resolve it: a feasible check, more information, or a narrower claim.
3. **Suggestions and minor comments**, in a separate section with its own numbering and labelled as optional: units, sample sizes that do not match across tables, missing standard errors, unclear definitions, typos.
4. **Overall assessment**: the main strength, the main limitation, what a revision must do.

Separate sections and numbering for essential problems and suggestions: Berk et al. (2017, pp. 234 and 241).

The letter to the editor

- Read by the editor only. According to Berk et al. (2017, pp. 239–240) it should be:
  - brief, not a copy of the report;
  - a **recommendation**: accept, revise, reject. “Be decisive”;
  - how important the contribution is relative to existing work, whether the analysis is convincing, and whether the paper is publishable now or within one round of revision;
  - consistent with the report: a glowing report with a “reject” in the letter confuses editor and authors.
- Also useful: what you could and could not check, and what a revision must show for you to support publication.
- Where does the recommendation go? Some journals want it only in the letter. Thomson (2011, p. 159) disagrees: when a paper is turned down, the authors are entitled to know the basis of the decision. Follow the journal’s instructions.
- If you used tools such as AI to prepare the report, say so, as the journal and your university require.

Rules of thumb

- Write the summary in your own words, not the abstract again (Thomson, p. 152).
- Number every request; do not lump several points together (Thomson, p. 158; Berk et al., p. 241).
- Separate statements of fact from judgments (Thomson, p. 160): "the table reports $N = 61$" is a fact; "this change of sample makes the comparison uninformative" is a judgment.
- Weigh the cost of every request: "The benefits must exceed the costs", and the authors bear the costs (Berk et al., p. 237).
- The paper is the authors'; their name goes on it. Do not ask for a different paper (Berk et al., pp. 236–237).
- Be tough, not mean (Thomson, pp. 161–162); do not ascribe bad intent to the authors (Berk et al., p. 242).

Thomson (2011), Chapter 4; Berk, Harvey and Hirshleifer (2017).

DISCUSS Good or weak?

Five comments on Voigt and Wulf (2019):

1. "The sample is too small to draw any conclusion."
2. "In Table 2, column (1), press freedom is reported as 0.01 (0.01) but marked significant at 5%. Please report three decimals so that readers can recover the t statistic."
3. "The authors should use a better identification strategy."
4. "Section 8 states that a 10-point increase in press freedom 'leads to' a 0.10-point rise in de facto independence. Press freedom is not randomly assigned and correlates 0.71 with GDP per capita (footnote 11). The sentence should describe an association, or the paper should explain which variation identifies an effect."
5. "I would have studied judges rather than prosecutors."

Classify each as good or weak. Rewrite one weak comment so that it becomes useful.

ANSWERS Good or weak?

- (1) Weak: no table, no number, no consequence. Useful version: "Column (5) uses 51 of the 74 countries of column (1). Please list the excluded countries and show column (1) on the same 51."
- (2) Good **minor** comment: specific, verifiable, easy to fix.
- (3) Weak: which variation, which problem, which alternative?
- (4) Good **main** comment: cites the text and a number, explains why it matters, offers two ways out.
- (5) Weak: it asks for a different paper. Judge the paper the authors wrote: their name goes on it (Berk et al., p. 237).

Your report for this course

Required in the syllabus	Part of the report
Accurate summary: question, contribution, data, unit, sample, method, findings	Summary
One central result explained: exact table or slide, magnitude, units, comparison, sample size, uncertainty	End of the summary, or first main comment
Two or three specific comments: issue, why it matters, feasible check	Main comments (plus minor ones if useful)
Overall assessment: a strength and the main limitation	Overall assessment

- No recommendation and no letter to the editor are required.

Laboratory

- Worksheet: `do/exercises/05_multiple_regression.do`.
- Data: `justice_europe_2019.dta` (Tasks 1–3) and `contracts_2019.dta` (Tasks 4–5).
- Task 6: write one main comment and one minor comment on Table 2 of Voigt and Wulf, with the structure of today's slides.
- Every coefficient as a sentence: unit, comparison group, what is held fixed.

Exit check

1. Write the police coefficient of the multiple regression (-2.04) as a sentence.
2. The simple coefficient is -3.64. Explain in one sentence why it is more negative.
3. In `regress time_days i.income4`, what does the constant measure?
4. Name the four parts of a referee report.

Next lecture

- Lecture 6 (Friday): when can we trust a regression's standard errors? Outliers and changing samples.
- Read Voigt and Wulf (2019): Section 8 in full, all columns of Table 2 and the footnotes.
- Today: hand in your one-page outline (question, data, method).